

# **Pizza Sales Data Analytics Project**

**Using Mysql**

# Project Overview:

- Analyzed pizza sales data for 2015 to understand customer ordering patterns and business performance.
- **Combined three datasets:**
  - Orders – order date and time
  - Order Details – pizzas ordered and quantities
  - Pizza Types – pizza names, categories, and ingredients
- Performed data cleaning, transformation, and analysis to identify key sales trends.
- Developed an interactive dashboard to present sales performance visually.
- The analysis focuses on popular pizzas, categories, order trends, peak hours, and customer purchasing behavior.

# Data Summary:

- **Dataset Overview:**
- **DatasetRecordsKey Information Orders = 21,350**
- **Order ID, Date, Time Order Details = 48,620**
- **Pizza ID, Quantity Pizza Types = 32**
- **Overall Data:**
- **21,350 unique orders**
- **48,620 order line items**
- **49,574 pizzas sold**
- **32 different pizza types**
- **4 product categories**
- **Analysis period: January–December 2015**

# Q1. Retrieve the total number of orders placed.

## Solution:

```
-- Retrieve the total number of orders places.  
SELECT  
    COUNT(order_id) AS total_orders  
FROM  
    orders;
```

## Result:

| Result Grid |              |
|-------------|--------------|
|             | total_orders |
| ▶           | 21350        |

## Q2. Calculate the total revenue generated from pizza sales.

### Solution:

```
-- Calculate the total revenue generated from pizza sales.alter

SELECT
  round(sum(orders_details.quantity * pizzas.price),2)AS total_sales
FROM
  orders_details
  JOIN
  pizzas ON pizzas.pizza_id = orders_details.pizza_id;
```

### Result:

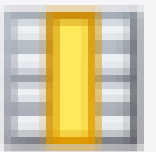
| Result Grid |             |
|-------------|-------------|
|             | total_sales |
| ▶           | 83474.9     |

# Q3. Identify the highest-priced pizza.

**Solution:**

```
-- Identify the highest- pricedpizza.  
  
SELECT  
    pizza_types.name, pizzas.price  
FROM  
    pizza_types  
    JOIN  
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id  
ORDER BY pizzas.price DESC  
LIMIT 1;
```

**Result:**

| Result Grid     Filter Rows: |                 |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------|
|                                                                                                                                                                                                        | name            | price |
| ▶                                                                                                                                                                                                      | The Greek Pizza | 35.95 |

# Q4. Identify the most common pizza size ordered.

**Solution:**

```
-- Identify the most common pizza size ordered.
```

```
SELECT
```

```
    pizzas.size,
```

```
    COUNT(orders_details.order_details_id) AS order_count
```

```
FROM
```

```
    pizzas
```

```
    JOIN
```

```
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
```

```
GROUP BY pizzas.size
```

```
ORDER BY order_count DESC;
```

**Result:**

| Result Grid |      |             |  | Filter |
|-------------|------|-------------|--|--------|
|             | size | order_count |  |        |
| ▶           | L    | 1909        |  |        |
|             | M    | 1546        |  |        |
|             | S    | 1461        |  |        |
|             | XL   | 56          |  |        |
|             | XXL  | 2           |  |        |

Q5. List the top 5 most ordered pizza types along with their quantities.

Solution:

```
SELECT
    pizza_types.name, SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result:

| Result Grid |                              |          |  |  Filter Rows: <input type="text"/> |
|-------------|------------------------------|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|             | name                         | quantity |                                                                                       |                                                                                                                         |
| ▶           | The Pepperoni Pizza          | 286      |                                                                                       |                                                                                                                         |
|             | The Barbecue Chicken Pizza   | 254      |                                                                                       |                                                                                                                         |
|             | The California Chicken Pizza | 245      |                                                                                       |                                                                                                                         |
|             | The Classic Deluxe Pizza     | 228      |                                                                                       |                                                                                                                         |
|             | The Hawaiian Pizza           | 225      |                                                                                       |                                                                                                                         |

**Q6. Join the necessary tables to find the total quantity of each pizza category ordered.**

**Solution:**

```
-- join the necessary tables to find the total quantity of
-- each pizza category ordered.
SELECT
    pizza_types.category,
    SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

**result:**

| Result Grid |          |          |  |  | File |
|-------------|----------|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|
|             | category | quantity |                                                                                     |                                                                                     |      |
| ▶           | Classic  | 14888    |                                                                                     |                                                                                     |      |
|             | Supreme  | 11987    |                                                                                     |                                                                                     |      |
|             | Veggie   | 11649    |                                                                                     |                                                                                     |      |
|             | Chicken  | 11050    |                                                                                     |                                                                                     |      |

**Q7. Determine the distribution of orders by hour of the day.**

**Solution:**

```
-- Determine the distribution of orders by hour of the day.  
  
SELECT  
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count  
FROM  
    orders  
GROUP BY HOUR(order_time);
```

**Result:**

| Result Grid |      |             |  | F |
|-------------|------|-------------|--|---|
|             | hour | order_count |  |   |
| ▶           | 11   | 1231        |  |   |
|             | 12   | 2520        |  |   |
|             | 13   | 2455        |  |   |
|             | 14   | 1472        |  |   |
|             | 15   | 1468        |  |   |
|             | 16   | 1920        |  |   |
|             | 17   | 2336        |  |   |
|             | 18   | 2399        |  |   |
|             | 19   | 2009        |  |   |
|             | 20   | 1642        |  |   |
|             | 21   | 1198        |  |   |
|             | 22   | 663         |  |   |

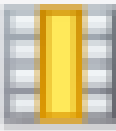
## Q8. Join relevant tables to find the category-wise distribution of pizzas.

### Solution:

```
-- join relevant table to find the category-wise  
-- distribution of pizzas.
```

```
SELECT  
    category, COUNT(name)  
FROM  
    pizza_types  
GROUP BY category;
```

### Result:

| Result Grid |          |             |  |  Filter Rows: |
|-------------|----------|-------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|             | category | COUNT(name) |                                                                                     |                                                                                                  |
| ▶           | Chicken  | 6           |                                                                                     |                                                                                                  |
|             | Classic  | 8           |                                                                                     |                                                                                                  |
|             | Supreme  | 9           |                                                                                     |                                                                                                  |
|             | Veggie   | 9           |                                                                                     |                                                                                                  |

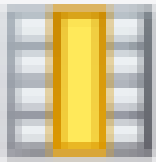
**Q9. Group the orders by date and calculate the average number of pizzas ordered per day.**

**Solution:**

```
-- Group the orders by date and calculate the average number of
-- pizzas ordered per day.

SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(orders_details.quantity) AS quantity
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

**Result:**

| Result Grid                                                                         |                |  |  | Filter |
|-------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|
|                                                                                     | Average_Pizzas |                                                                                       |                                                                                       |        |
|  | 138            |                                                                                       |                                                                                       |        |

**Q10. Determine the top 3 most ordered pizza types based on revenue.**

**Solution:**

```
-- Determine the top 3 most ordered pizza types based
-- on revenue for each pizza category
select name, revenue from
) (select category, name, revenue,
rank() over (partition by category order by revenue desc) as rn from
) (select pizza_types.category, pizza_types.name,
sum((orders_details.quantity) * pizzas.price) as revenue
from pizza_types join pizzas on
pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <=3;
```

**Result:**

| Result Grid   Filter Rows: <input type="text"/> |                              |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------|
|                                                                                                                                                                                                                         | name                         | revenue  |
| ►                                                                                                                                                                                                                       | The Thai Chicken Pizza       | 43434.25 |
|                                                                                                                                                                                                                         | The Barbecue Chicken Pizza   | 42768    |
|                                                                                                                                                                                                                         | The California Chicken Pizza | 41409.5  |
|                                                                                                                                                                                                                         | The Classic Deluxe Pizza     | 38180.5  |
|                                                                                                                                                                                                                         | The Hawaiian Pizza           | 32273.25 |
|                                                                                                                                                                                                                         | The Pepperoni Pizza          | 30161.75 |

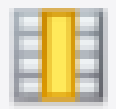
**Q11. Calculate the percentage contribution of each pizza type to total revenue.**

**Solution:**

```
-- Calculate the percentage contribution of each
-- pizza type to total revenue.

select pizza_types.category, round(sum(orders_details.quantity * pizzas.price)
/ (select round(sum(orders_details.quantity * pizzas.price),2)as total_sales
from orders_details join pizzas
on pizzas.pizza_id = orders_details.pizza_id) *100,2 )as revenue
from pizza_types join pizzas on pizza_types.pizza_type_id =
pizzas.pizza_type_id join orders_details on orders_details.pizza_id =
pizzas.pizza_id group by pizza_types.category order by revenue desc;
```

**Result:**

| Result Grid |          |         |  |  | Filter |
|-------------|----------|---------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|
|             | category | revenue |                                                                                       |                                                                                       |        |
| ▶           | Classic  | 26.91   |                                                                                       |                                                                                       |        |
|             | Supreme  | 25.46   |                                                                                       |                                                                                       |        |
|             | Chicken  | 23.96   |                                                                                       |                                                                                       |        |
|             | Veggie   | 23.68   |                                                                                       |                                                                                       |        |

## Solution:

```
-- Analyze the cumulative revenue generated over time.
select order_date, sum(revenue)
over (order by order_date) as cum_revenue
from (select orders.order_date,
sum(orders_details.quantity * pizzas.price)
as revenue from orders_details join pizzas
on orders_details.pizza_id = pizzas.pizza_id
join orders on orders.order_id = orders_details.order_id
group by orders.order_id) as sales;
```

# Result:

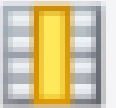
[illegible]

Q13. Determine the top 3 most ordered pizza types based on revenue for each pizza category.

## Solution:

```
-- Determine the top 3 most ordered
-- pizza types based on revenue.
SELECT pizza_types.name,
SUM(orders_details.quantity * pizzas.price) AS revenue
FROM pizza_types JOIN pizzas
ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

## Result:

| Result Grid     Filter Rows: <input type="text"/> |                              |          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------|
|                                                                                                                                                                                                                             | name                         | revenue  |
| ▶                                                                                                                                                                                                                           | The Thai Chicken Pizza       | 43434.25 |
|                                                                                                                                                                                                                             | The Barbecue Chicken Pizza   | 42768    |
|                                                                                                                                                                                                                             | The California Chicken Pizza | 41409.5  |

# THANK YOU

